

JEE MAIN REVISION MASTER: POLYMERS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 15 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Polymers — Core Principles & Governing Laws	13 PYQs	Core Principle
Standard	Polymers — Standard Calculations & Problem Archetypes	1 PYQs	Quantitative Analysis
Standard	Polymers — Applied Cases, Reactions & Traps	1 PYQs	Applied Topics

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Polymers — Core Principles & Governing Laws	$[H_{2}N-CH_{2}-COOH]$, $(CH_{3})_{3}SiCl$, $[HOOC-(CH_{2})_{4}COOH]$
Polymers — Applied Cases, Reactions & Traps	$(C_{2}H_{5})_{3}Al$, $TiCl_{4}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Polymers — Core Principles & Governing Laws (High Yield — 13 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Polymers.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Polymers.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Polymers PYQs.

Representative Source PYQs:

- [2024 Shift] Q#1 (MCQ): "Which of the following set of polymers are used as fibre? (i) Teflon (ii) Starch (iii) Terylene (iv) Orlon..." [Ans: N/A]
- [2024 Shift] Q#2 (MCQ): "The biodegradable polymer obtained by polymerisation of glycine and aminocaproic acid is..." [Ans: N/A]
- [2023 Shift] Q#3 (MCQ): "Which of the following monomers form biodegradable polymers ?..." [Ans: N/A]

3.2 Polymers — Standard Calculations & Problem Archetypes (Standard Yield — 1 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Polymers.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2017 Shift] Q#3 (MCQ): "The monomer used in novolac, a polymer used in paints..." [Ans: N/A]

3.3 Polymers — Applied Cases, Reactions & Traps (Standard Yield — 1 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Polymers.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2018 Shift] Q#1 (MCQ): "Ziegler-Natta catalyst is used to prepare..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2024 (Q#1)

Which of the following set of polymers are used as fibre? (i) Teflon (ii) Starch (iii) Terylene (iv) Orlon

- (A) (i) and (ii)
- (B) (ii) and (iii)
- (C) (iii) and (iv)
- (D) (i) and (iv)

Correct Answer: None

Solution Analysis: The polymers that are used as fibers are : Terylene : Also known as polyester, this is a synthetic polymer that is widely used as a fiber in textiles due to its durability, resistance to stretching and shrinking, and quick-drying properties. Orlon : This is a trade name for polyacrylonitrile, a synthetic polymer that is used to make acrylic fibers, known for their wool-like feel and used for sweaters, hats, and other garments. Therefore, the correct option is : Option C : (iii) and (iv)

Benchmark Problem 2: KCET 2018 (Q#1)

Ziegler-Natta catalyst is used to prepare

- (A) low-density polythene
- (B) teflon
- (C) high density polythene
- (D) nylon-6

Correct Answer: None

Solution Analysis: Ziegler-Natta catalyst is composed of TiCl_4 and $(\text{C}_2\text{H}_5)_3\text{Al}$. This catalyst is specifically used to produce high-density polyethylene (HDPE). The polymerization reaction proceeds as follows: Ethene (C_2H_4) $\xrightarrow{\text{Ziegler-Natta catalyst}}$ Polythene (HDPE). In this reaction, ethene monomers ($\text{CH}_2=\text{CH}_2$) are polymerized under the influence of the Ziegler-Natta catalyst and specific conditions of temperature (333-343 K) and pressure (6-7 atm) to form high-density polyethylene chains $(\text{CH}_2-\text{CH}_2)_n$.

Benchmark Problem 3: KCET 2017 (Q#3)

The monomer used in novolac, a polymer used in paints

- (A) melamine and formaldehyde
- (B) phenol and formaldehyde
- (C) butadiene and acrylo nitrile
- (D) butadiene and styrene

Correct Answer: None

Solution Analysis: Novolacs are phenol formaldehyde resins with formaldehyde to phenol molar ratio of less than one. The polymerisation is brought to completion using acid catalysis such as oxalic acid, hydrochloric acid or sulphonate.